Scatterplots and Correlation

APHG 20293

Does More Sleep Go With Less Stress?

Research Question

- Imagine a campus wellness team asks 20 students about sleep and stress during the same week.
- **Research Question:** Do students who report more sleep also report less stress?

Consider

- What data would we want to begin to address this research question?

One Row, One Student

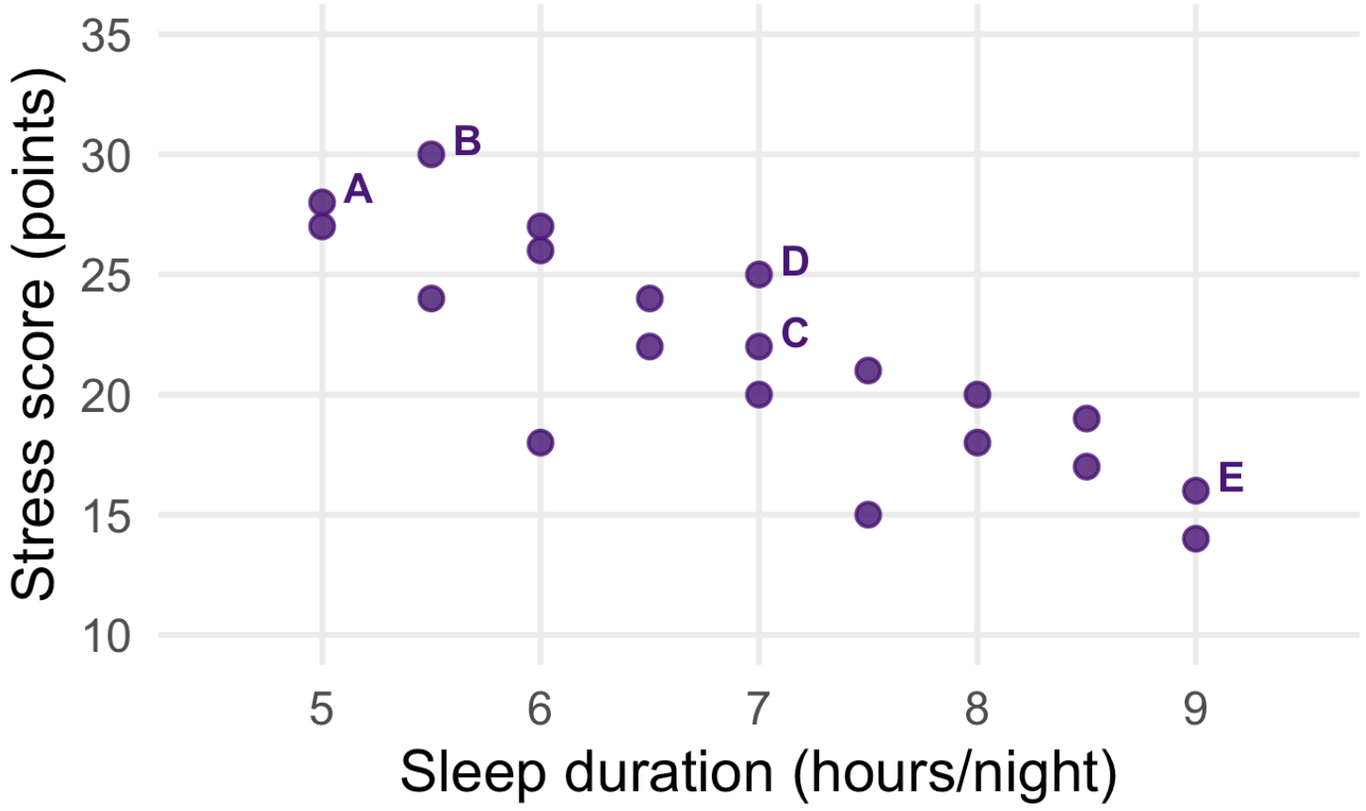
Student	Sleep (hours/night)	Stress (points)
S01	5.0	28
S02	7.0	22
S03	6.0	26
S04	8.0	18

Questions to Consider

- What numerical or graphical summaries could we use to investigate whether sleep and stress are associated?
- What would we expect to see if sleep and stress are associated?

Graphical Summaries of Two Numerical Variables: Scatterplots

Summarize with a Scatterplot



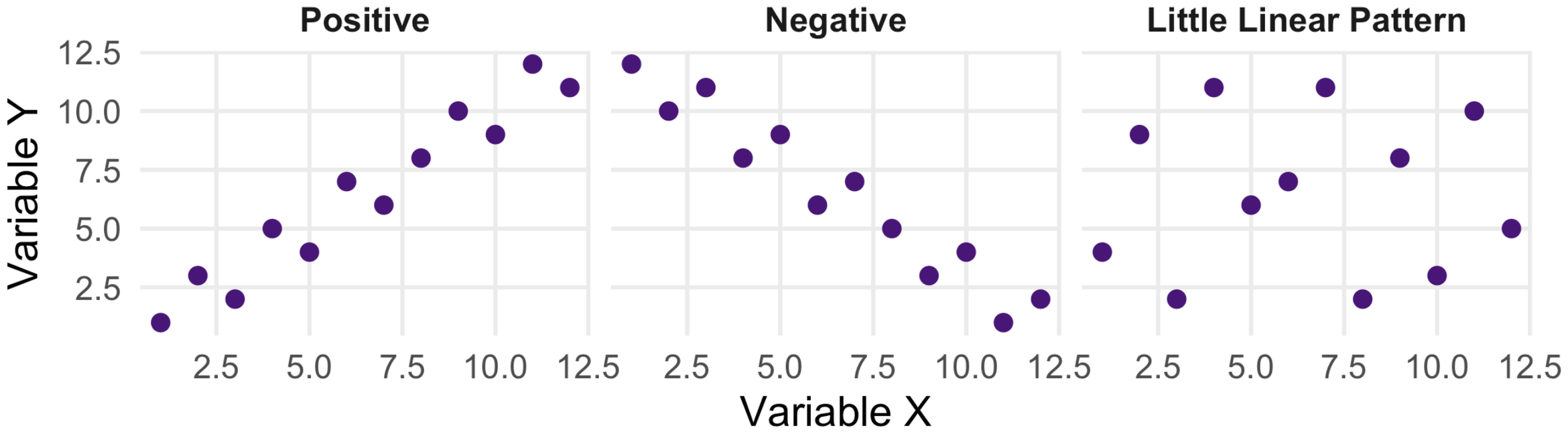
Each point is **one student**.

- **Student S01:**
  - Sleep = 5.0
  - Stress = 28
- **Student S02:**
  - Sleep = 7.0
  - Stress = 22

Four Things to Look For

1. **Direction:** upward, downward, or neither?
2. **Strength:** how closely do points follow a pattern?
3. **Form:** approximately straight or curved?
4. **Unusual observations:** does a point stand apart?

Direction of the Relationship



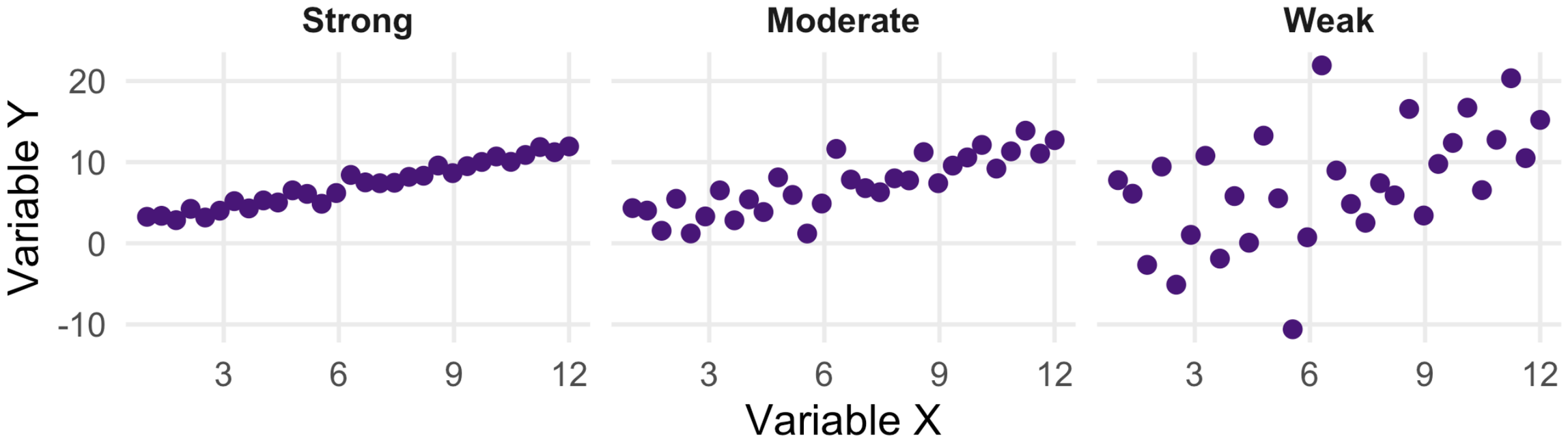
Positive: higher X tends to accompany higher Y.

Negative: higher X tends to accompany lower Y.

Four Things to Look For

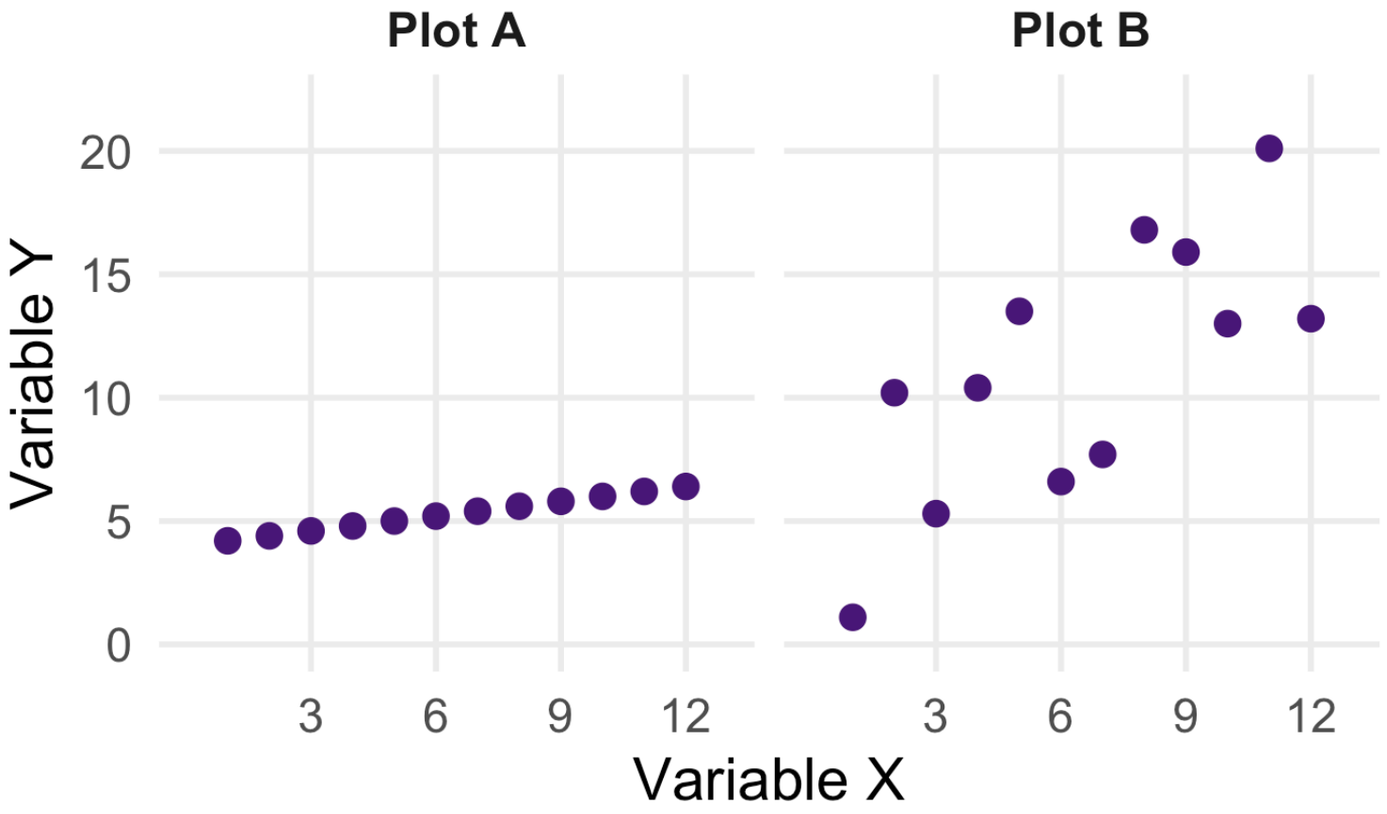
1. **Direction:** upward, downward, or neither?
2. **Strength:** how closely do points follow a pattern?

Strength of the Relationship



All three have the **same underlying slope**. They differ in how tightly the points cluster around it.

Question: Which Relationship Is Stronger?



Which shows a **stronger linear relationship**?

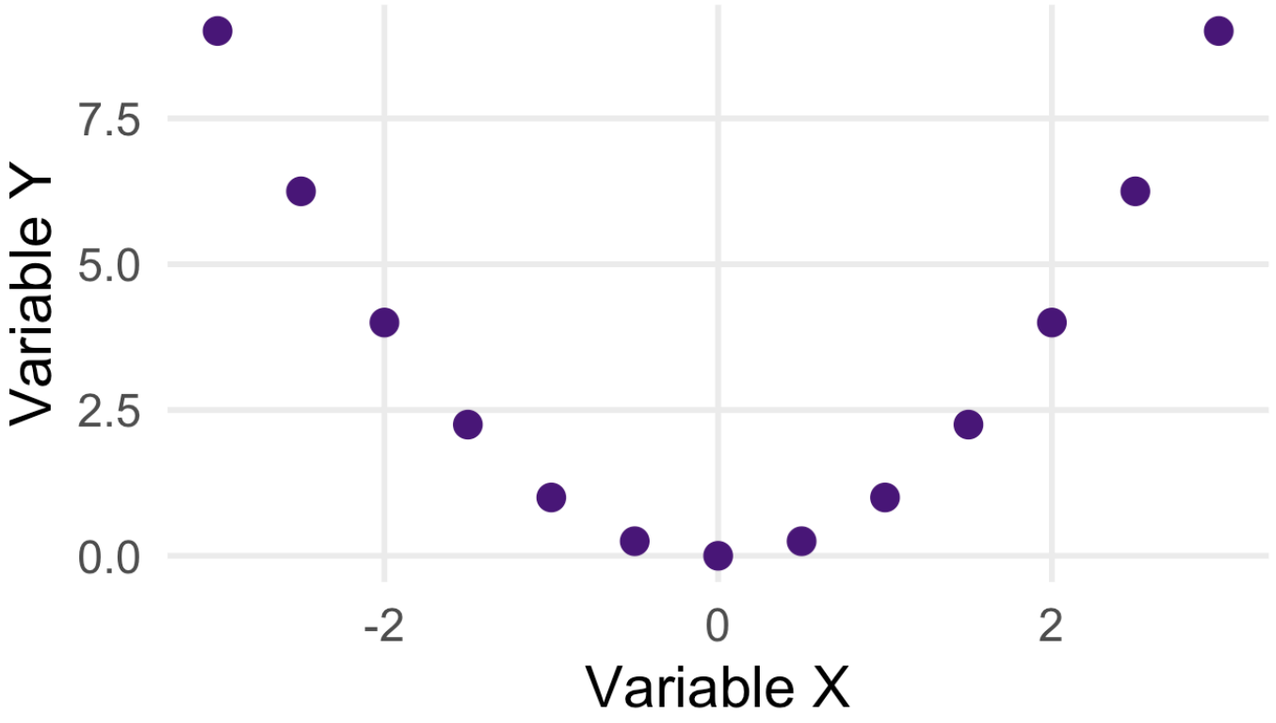
A. Plot A
B. Plot B
C. Both are equally strong
D. Cannot tell from the plots

Choose one and explain why.

Four Things to Look For

1. **Direction:** upward, downward, or neither?
2. **Strength:** how closely do points follow a pattern?
3. **Form:** approximately straight or curved?

A Relationship Can Be Nonlinear

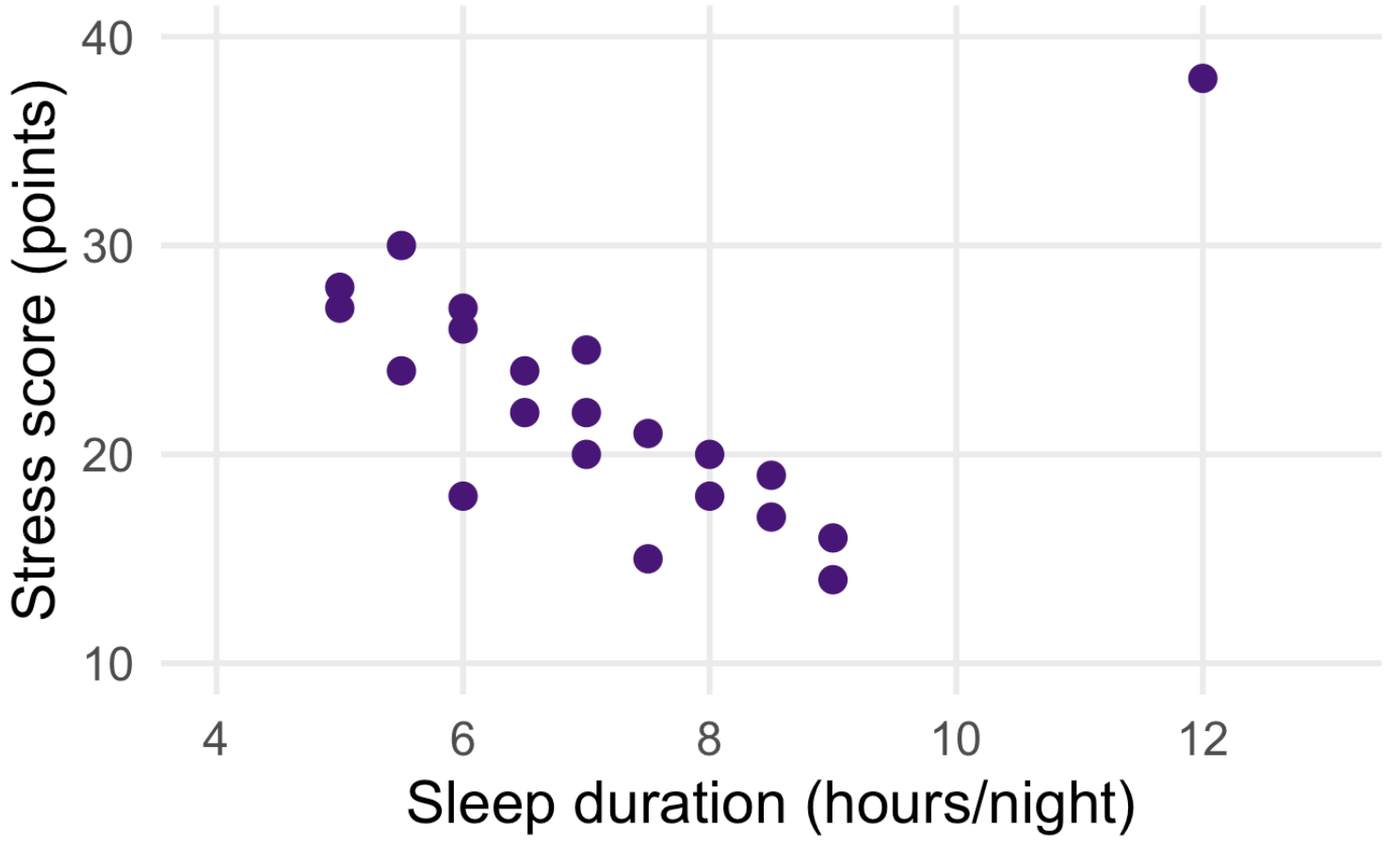


This U-shaped pattern is one example of a nonlinear relationship. Other patterns may rise and then fall, or level off.

Four Things to Look For

1. **Direction:** upward, downward, or neither?
2. **Strength:** how closely do points follow a pattern?
3. **Form:** approximately straight or curved?
4. **Unusual observations:** does a point stand apart?

Unusual Observations



- Which observation stands apart from the overall pattern?
- What would you want to check?

Four Things to Look For

1. **Direction:** upward, downward, or neither?
2. **Strength:** how closely do points follow a pattern?
3. **Form:** approximately straight or curved?
4. **Unusual observations:** does a point stand apart?

Numerical Summaries of Two Numerical Variables

Pearson's Correlation Coefficient

r ranges from –1 to +1.

Feature	Interpretation
Sign	Direction of the linear relationship
Absolute value	Strength of the linear relationship
Near –1 or +1	Points lie close to a straight line
Near 0	Little linear relationship

Question: Which Relationship Is Strongest?

Consider these correlation coefficients:

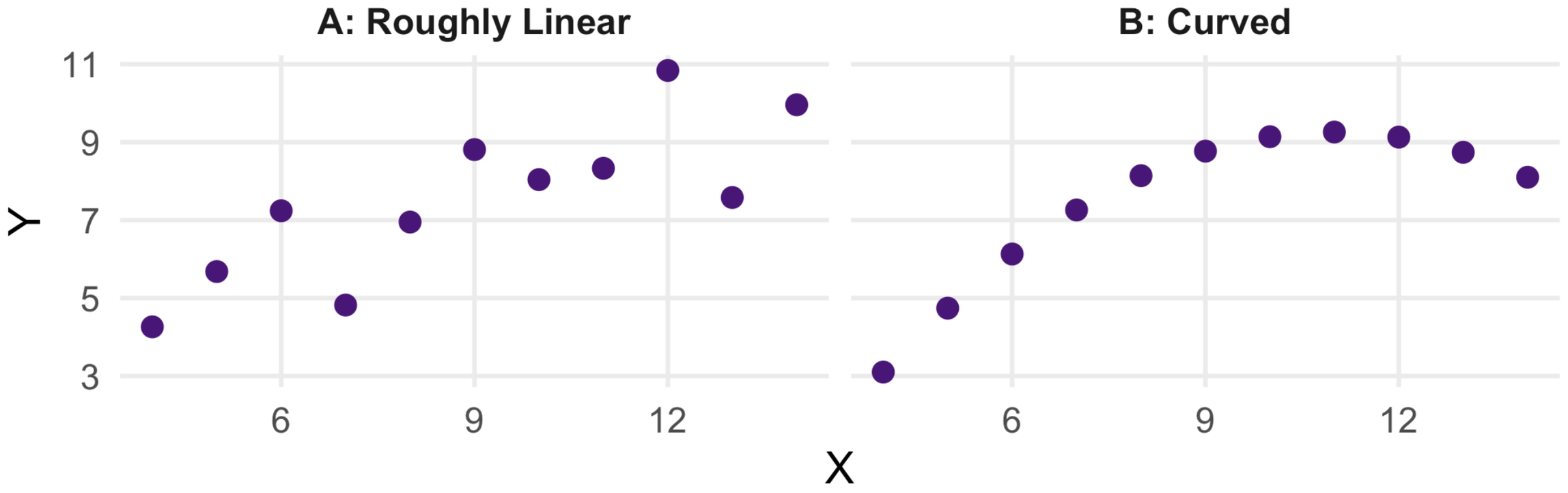
A. $r = +0.35$

B. $r = -0.80$

C. $r = +0.60$

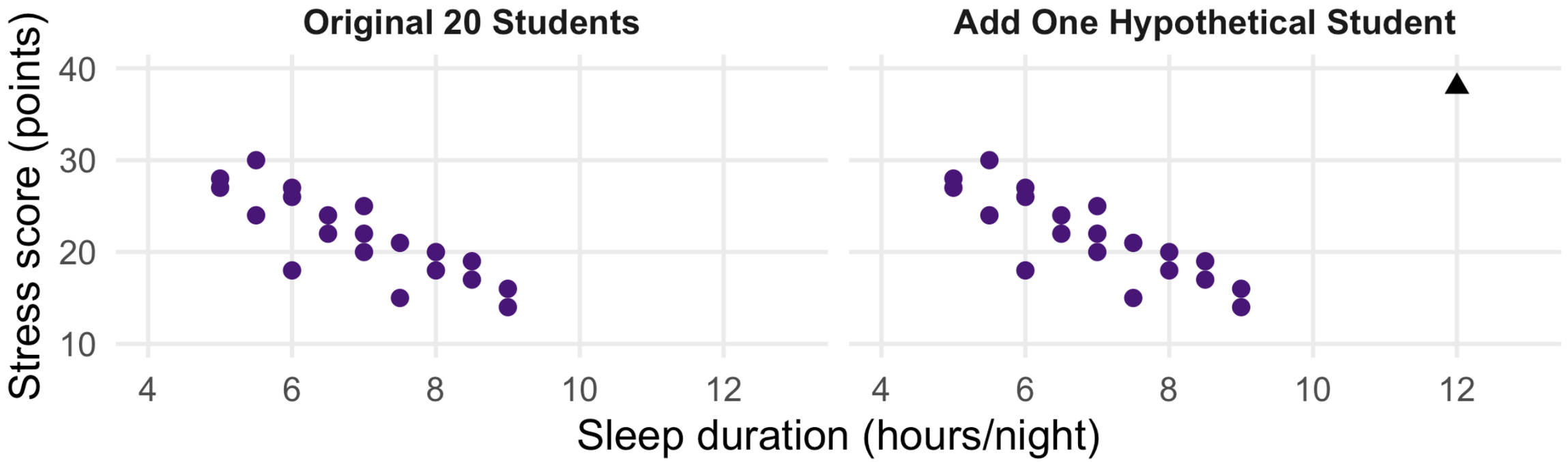
D. $r = -0.10$

Similar r, Different Patterns



Both have **r ≈ 0.816**. A single coefficient does not show the shape.

One Observation Can Change r



Original r = **-0.834**. With the added observation, r = **-0.077**.

Read a Correlation Matrix

	Sleep	Stress	Screen Time
Sleep	1.00	-0.83	-0.70
Stress	-0.83	1.00	0.86
Screen Time	-0.70	0.86	1.00

- Find the intersection of the two variables.
- The diagonal is 1.00.
- Values repeat across the diagonal.

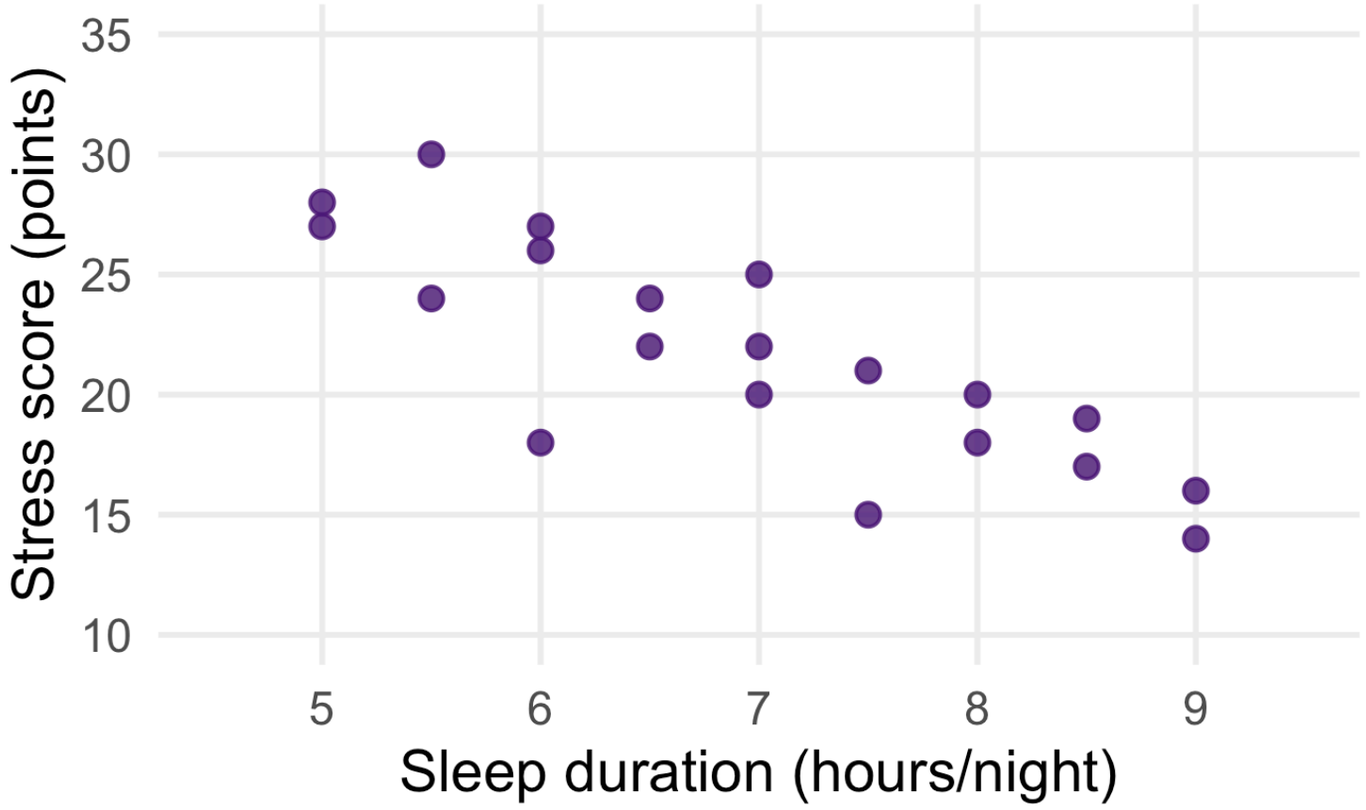
Compare the Correlations

	Sleep	Stress	Screen Time
Sleep	1.00	-0.83	-0.70
Stress	-0.83	1.00	0.86
Screen Time	-0.70	0.86	1.00

In these data, which pair has the strongest linear relationship, and in what direction?

Interpreting and Applying Correlation

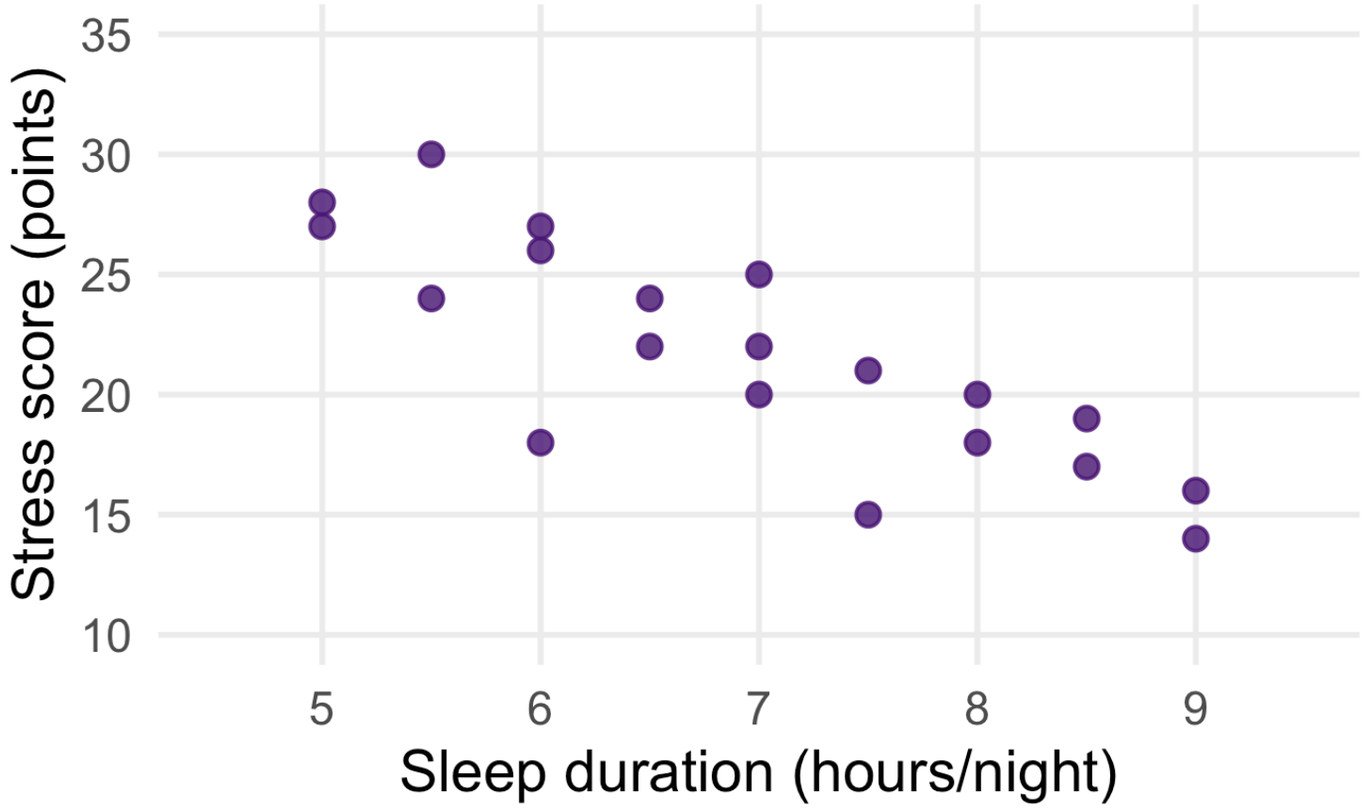
Correlation and Causation



Students who sleep more might report less stress because:

- Sleep affects stress.
- Stress affects sleep.
- Workload or another factor affects both.

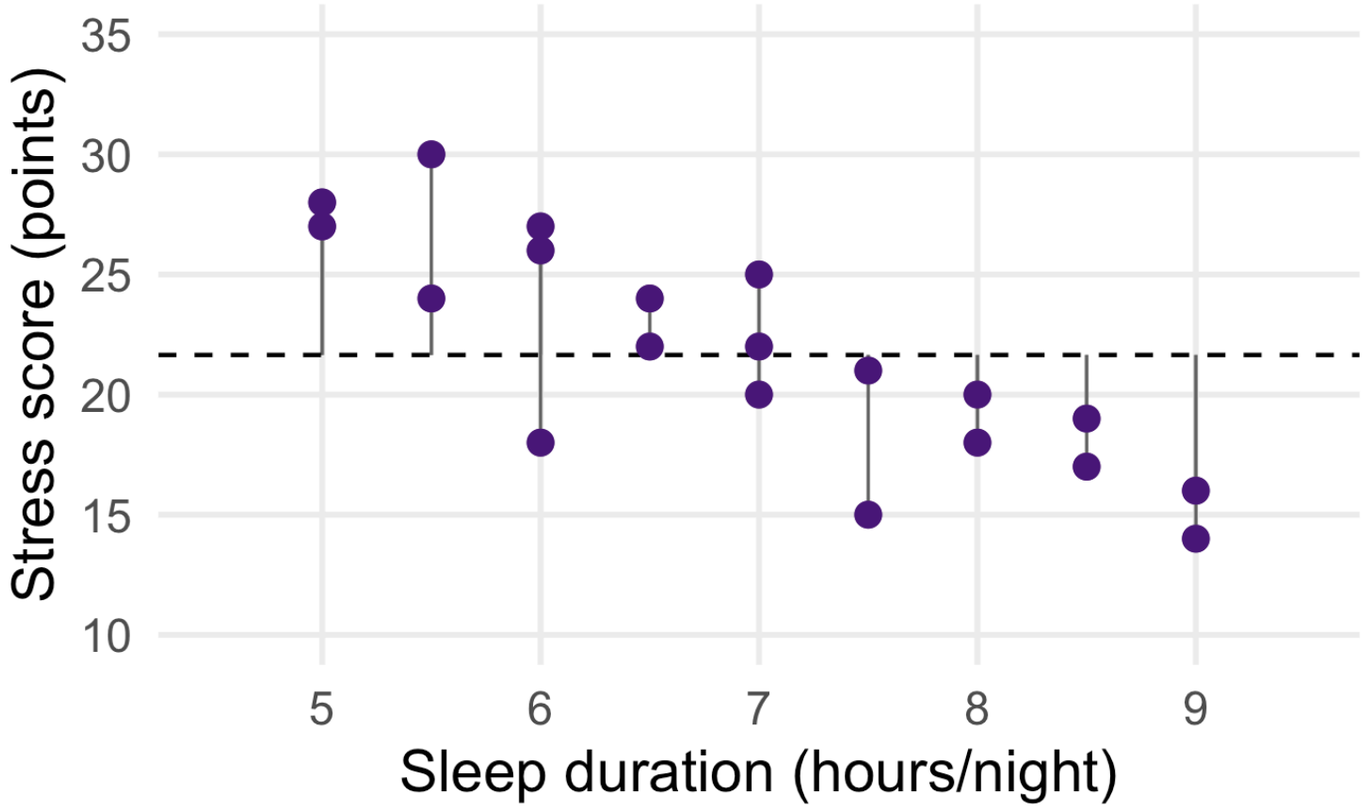
Predicting Stress



Now, suppose we want to predict a student's stress score.

Without knowledge of their sleep duration, what is our best prediction?

Predicting Stress With the Mean

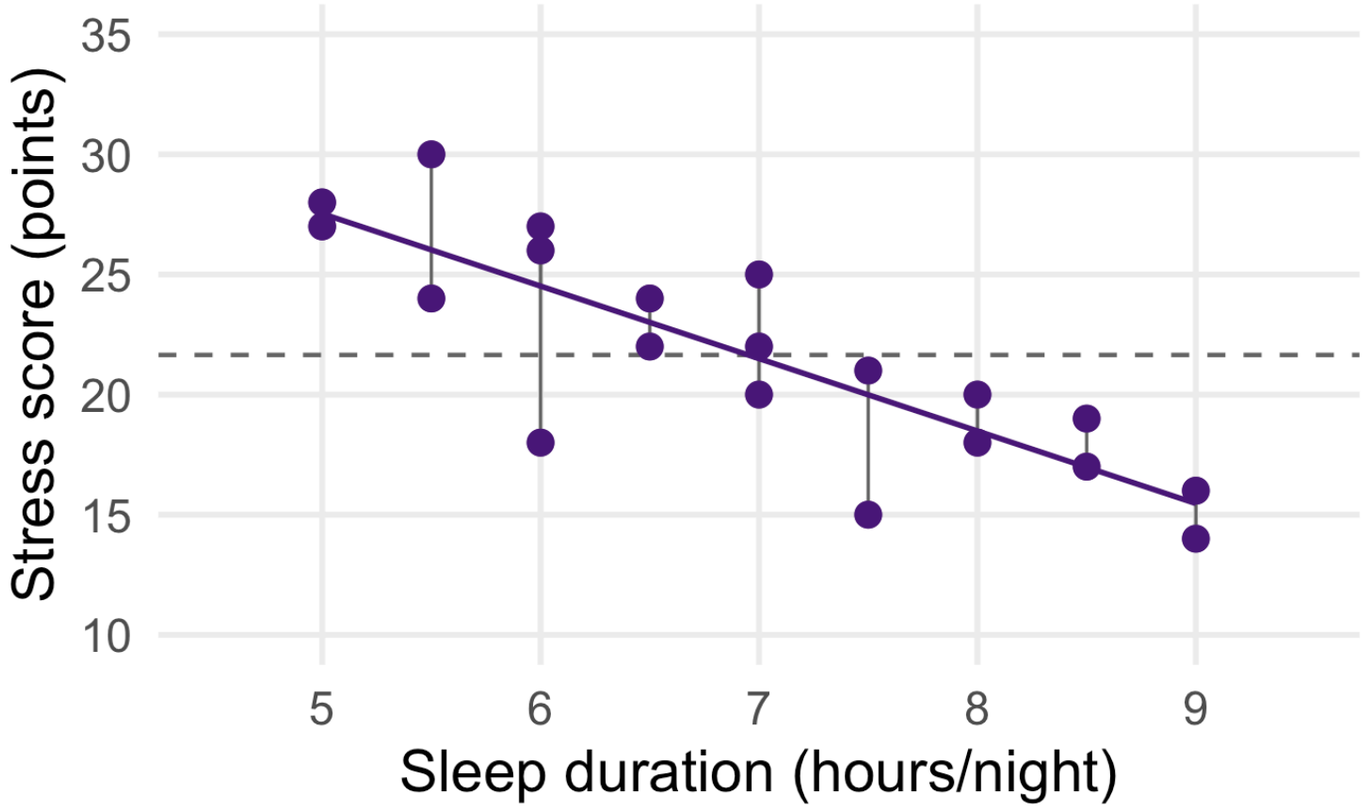


Without using sleep, our best prediction is the **mean stress score** for everyone:

21.65 points.

Each vertical segment shows an observed score's distance from that prediction.

Predicting Stress With Sleep



The **regression line** predicts stress using sleep duration.

The vertical segments now show errors relative to the line.

From r to r^2

r^2, the coefficient of determination, measures the proportion of variation accounted for by the regression line.

Prediction	Sum of Squared Errors
Mean only	404.55
Regression line	122.99

$$r^2 = 1 - \frac{122.99}{404.55} \approx 0.696$$

Using sleep reduces the sum of squared errors by **69.6%** compared with using the mean alone.

For simple linear regression with an intercept, this also equals **Pearson's r squared**.

Why We Can Square r

For **ordinary least-squares regression**:

- **Simple** means one predictor: sleep duration.
- **With an intercept** means the line can shift up or down instead of being forced through (0, 0).

Under these conditions, both calculations give the same r^2:

Square the correlation

$$(-0.834253)^2 \approx 0.696$$

Compare squared errors

$$1 - \frac{122.99}{404.55} \approx 0.696$$

r retains direction. **r^2** describes the proportion of variation accounted for by the line.

Open the Practice Workbook

Download correlation_practice.xlsx and open **Practice**.

Column	Contents
A	Student identifier
B	Sleep duration, hours/night
C	Fictional stress score, points
D	Screen time, hours/day

Use **B and C** today. Data occupy **rows 2–21**.

Select the Paired Data

1. Select **B1:C21**, including both headers.
2. Keep **sleep_hours on the left** and **stress_score on the right**.
3. Exclude the student identifier, screen time, and calculation cells.

One row supplies one X-Y pair. Keep those rows together.

Insert an XY Scatterplot

1. Choose **Insert → Scatter (X, Y)**.
2. Select **markers only**, with no connecting lines.
3. Move the chart into the open space below the instructions.

The horizontal axis should represent **sleep duration**. The observed values range from **5 to 9 hours**, but Excel may display a wider axis range, such as **0–10**.

Give the Chart Meaningful Labels

Use **Chart Design → Add Chart Element → Axis Titles**, or the chart's equivalent controls.

- **Title:** Sleep Duration and Stress Score.
- **Horizontal axis:** Sleep duration (hours/night).
- **Vertical axis:** Fictional stress score (points).

Check the point at **5 hours and 28 points** against the first data row.

Check the Graph Before Interpreting It

- Did Excel use **B2:B21 for X** and **C2:C21 for Y**?
- Are both ranges numeric and the same length?
- Does the downward pattern match the paired data?
- Are the title, labels, and points readable?

For Prep 3, a labeled scatterplot and visual description are enough.

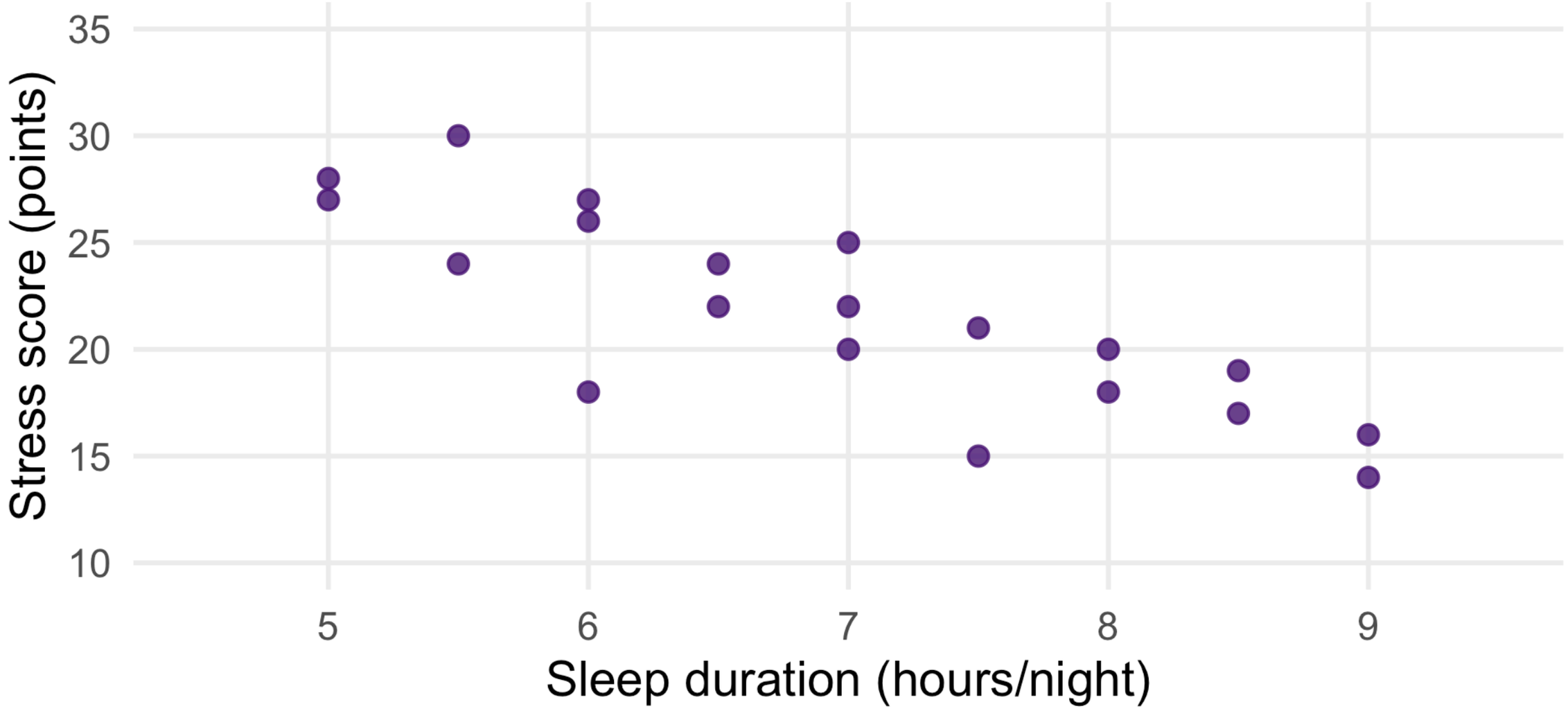
Group Exercise: Describe the Relationship

Work together for **8 minutes**:

1. Finish your scatterplot in **Practice**.
2. Write **2–3 bullets** about direction, apparent strength, and possible outliers.
3. Check whether the pattern is approximately linear.
4. Be ready to explain one point and one overall feature.

Choose an Excel operator, a range checker, and an interpreter.

Practice Graph



Compare Your Interpretations

- Students reporting more sleep generally report **lower stress scores**.
- The downward pattern is **fairly strong and approximately linear**, with some scatter.
- The point at **6 hours and 18 points** sits below nearby observations and is worth checking.

We cannot tell from appearance alone whether a point is an error.

Calculate r in Excel

In **G3** on Practice, enter:

```
=CORREL(B2:B21,C2:C21)
```

Result: **r = -0.834**.

More sleep accompanies lower stress in this sample, with a fairly strong **negative linear relationship**.

Calculate r^2 and Check n

Cell	Formula	Result
G4	`=G3^2`	0.696
G5	`=COUNT(B2:B21)`	20

About **69.6%** of the variation in stress scores is accounted for by the linear relationship with sleep.

Compare With the Answer Key

Answer Key contains:

- The original practice observations.
- An editable scatterplot linked to its data.
- Formulas for r, r^2, and the number of observations.
- An example visual interpretation.

If your result differs, check **the ranges and row pairing first**.

Apply the Workflow to Prep 3

Use your **assigned dataset A–D**, with the assigned pair:

Groups	X: Correlation Predictor	Y: Continuous Outcome
1, 3, 5, 7	SleepHours	StressScore
2, 4, 6, 8	ScreenTimeHours	AnxietyScore

Create a labeled scatterplot and write **2–3 descriptive bullets**.

Prep 3 Submission Check

- Group number, dataset, names, and assigned-variable classifications.
- Outcome **n, mean, SD, minimum, and maximum**.
- Labeled histogram with **3–4 descriptive bullets**.
- Labeled scatterplot with **2–3 descriptive bullets**.
- Each member's **3–5-sentence contribution paragraph**.

One group document: **1–2 pages excluding graphs**, named `Group#_Prep3`.

Sources

- Yan Zhang. *Correlations and Reliability*, APHS 20203. Adapted correlation concepts and activities, PDF pages 3–21.
- APHS 20203. Approved remaining-course outline and *Group Presentation - Prep 3*.
- Microsoft Support: Scatter charts and CORREL.
- Penn State STAT 501: Coefficient of determination.
- F. J. Anscombe (1973). *Graphs in Statistical Analysis*. Datasets I and II supplied in R's `datasets::anscombe`.

The sleep, stress, and screen-time observations are **fictional teaching data**.